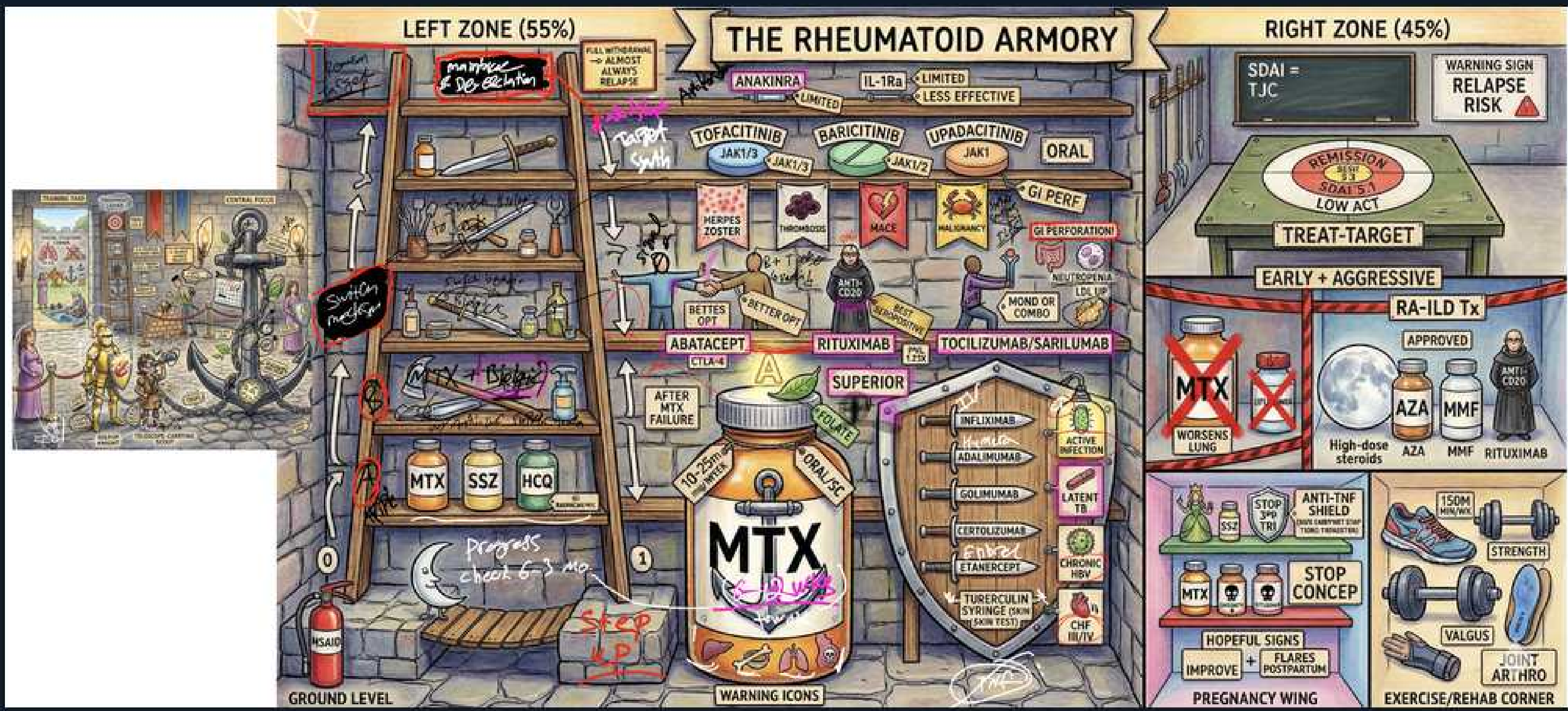
Rheumatoid Arthritis — Treatment & Updates



1. Methotrexate (anchor)

WHAT YOU SEE

Giant MTX bottle dominating the center as the foundation

WHAT IT ENCODES

Methotrexate is the ANCHOR first-line conventional synthetic DMARD for RA and should be started EARLY (within the window of opportunity) in essentially all patients without contraindication. It is dosed WEEKLY (typically 7.5–25 mg PO or SC weekly — never daily), works as an antifolate/anti-inflammatory by raising adenosine, and reduces both symptoms and radiographic erosion progression. Monitor CBC, LFTs, and renal function periodically; key toxicities are hepatotoxicity, myelosuppression, oral ulcers, and pneumonitis.

BOARD TRAP

The classic lethal error is methotrexate dosed DAILY instead of WEEKLY (causes fatal pancytopenia/mucositis). Another trap: expecting immediate relief — MTX takes ~6–8 weeks to work, so it is a disease-modifier, not a fast analgesic, which is why steroids are used to bridge. Choosing 'MTX for rapid pain control' is wrong.

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2. Folate supplementation

WHAT YOU SEE

FOLATE-shield warning icons clustered at the base of the giant MTX bottle — folic acid is the protective rescue co-prescribed with methotrexate to blunt its antifolate toxicities.

WHAT IT ENCODES

Folic acid (commonly 1 mg daily, or folinic acid weekly) is co-prescribed with methotrexate to reduce its antifolate side effects — stomatitis/mouth ulcers, GI intolerance (nausea), cytopenias, and hepatotoxicity — WITHOUT reducing efficacy. This supplementation is standard of care for every patient on MTX.

BOARD TRAP

A patient on MTX develops mouth ulcers, cytopenias, or transaminitis and the wrong answer is to stop MTX outright. The discriminator: ensure FOLIC ACID supplementation first (and check it isn't accidental daily MTX dosing); folate co-prescription mitigates these toxicities and lets the patient continue an effective DMARD.

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3. MTX contraindications

WHAT YOU SEE

STOP CONCEP / pregnancy wing icons

WHAT IT ENCODES

Methotrexate is ABSOLUTELY contraindicated in pregnancy and breastfeeding (it is a TERATOGEN and abortifacient — folate-antagonist neural tube/limb defects) and must be stopped and washed out BEFORE conception (≈3 months, in both women AND men planning pregnancy). It is also contraindicated in significant liver disease and heavy alcohol use (additive hepatotoxicity), and used cautiously in renal impairment (renally cleared — accumulation causes toxicity).

BOARD TRAP

A woman with RA on methotrexate who wants to conceive (or is pregnant) and the wrong answer continues MTX. The discriminator: MTX is an ABSOLUTE contraindication in pregnancy — switch to a pregnancy-compatible DMARD such as HYDROXYCHLOROQUINE or SULFASALAZINE (and stop leflunomide, which also requires cholestyramine washout). Continuing MTX peri-conception is the error.

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4. Conventional DMARDs (SSZ, HCQ, LEF)

WHAT YOU SEE

Ground level shelf: MTX, SSZ, HCQ

WHAT IT ENCODES

Other conventional synthetic DMARDs (csDMARDs) include SULFASALAZINE, HYDROXYCHLOROQUINE, and LEFLUNOMIDE. They are used as alternatives/add-ons when MTX is contraindicated or insufficient (e.g., 'triple therapy' = MTX + SSZ + HCQ). Sulfasalazine and hydroxychloroquine are the preferred DMARDs in PREGNANCY. Leflunomide (inhibits pyrimidine synthesis) is teratogenic and needs cholestyramine washout before conception.

BOARD TRAP

Picking leflunomide for a patient planning pregnancy (it is teratogenic, like MTX), or forgetting that SSZ/HCQ are the pregnancy-safe options. The discriminator: among csDMARDs, only sulfasalazine and hydroxychloroquine are pregnancy-compatible; methotrexate and leflunomide are contraindicated.

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5. Steroids = bridge only

WHAT YOU SEE

High-dose steroids X-marked / worsening shield on right target zone

WHAT IT ENCODES

Glucocorticoids (low-dose oral prednisone or intra-articular injections) are used as SHORT-TERM BRIDGING therapy to control inflammation rapidly while a DMARD takes its 6–8 weeks to work, and for acute flares. They act fast but are NOT a long-term monotherapy because of cumulative toxicity (osteoporosis, hyperglycemia/diabetes, hypertension, infection, weight gain, cataracts, avascular necrosis).

BOARD TRAP

A flaring RA patient maintained on chronic prednisone monotherapy, with the wrong answer being to continue/escalate steroids long-term. The discriminator: steroids BRIDGE only — they do not prevent erosions adequately long-term and their toxicity demands tapering off once DMARDs are effective. Long-term steroid monotherapy for RA is incorrect.

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6. Treat-to-target

WHAT YOU SEE

TREAT-TARGET dartboard with low-activity bullseye

WHAT IT ENCODES

RA is managed by a TREAT-TO-TARGET strategy: define an objective goal (clinical REMISSION, or at least LOW disease activity) measured with composite scores (e.g., DAS28, CDAI, SDAI), reassess every 1–3 months, and ESCALATE or switch therapy if the target isn't met. This structured, target-driven approach produces better outcomes than ad hoc adjustment.

BOARD TRAP

Leaving a patient on an unchanged regimen despite persistent moderate/high disease activity ('they feel okay'). The discriminator: treat-to-target mandates objective reassessment and treatment ESCALATION when remission/low activity isn't achieved — continuing a failing regimen without escalation violates the strategy and allows ongoing joint damage.

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7. Early & aggressive Rx

WHAT YOU SEE

EARLY + AGGRESSIVE banner above the target

WHAT IT ENCODES

There is a 'WINDOW OF OPPORTUNITY' early in RA during which prompt, aggressive DMARD therapy prevents irreversible cartilage/bone erosion and disability. Guidelines call for starting a DMARD (methotrexate) as soon as the diagnosis is made — delaying for symptomatic NSAID trials worsens long-term structural and functional outcomes.

BOARD TRAP

A newly diagnosed RA patient managed with NSAIDs and 'watchful waiting' before considering DMARDs. The discriminator: erosions are irreversible and occur early, so DMARDs must start at diagnosis — deferring disease-modifying therapy squanders the window of opportunity. Choosing symptomatic-only management for early RA is the trap.

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8. Anti-TNF biologics

WHAT YOU SEE

The lower armory shelf lined with the anti-TNF vials INFLIXIMAB, ADALIMUMAB, GOLIMUMAB, CERTOLIZUMAB and ETANERCEPT — the first biologic class, each weapon aimed at the central TNF cytokine driving synovitis.

WHAT IT ENCODES

TNF-alpha inhibitors — etanercept (soluble receptor), and the monoclonal antibodies adalimumab, infliximab, golimumab, certolizumab — are the usual first BIOLOGIC class, added when methotrexate (or csDMARD therapy) fails to reach target. They are typically combined WITH methotrexate for better efficacy and reduced anti-drug antibodies. They block the central pro-inflammatory cytokine driving synovitis and erosion.

BOARD TRAP

Starting an anti-TNF without infection screening. The discriminator: SCREEN for latent TUBERCULOSIS (IGRA/PPD + CXR) and hepatitis B/C BEFORE any anti-TNF — these agents can REACTIVATE latent TB and HBV. Also avoid TNF inhibitors in moderate-severe heart failure (worsen it) and in demyelinating disease. Skipping TB/HBV screening is the classic error.

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9. Rituximab

WHAT YOU SEE

Rituximab labeled bottle in mid shelf

WHAT IT ENCODES

Rituximab is an anti-CD20 monoclonal antibody that depletes B cells; in RA it is generally used AFTER anti-TNF failure (or as an alternative biologic), often with methotrexate. It is particularly favored when there is a contraindication to TNF inhibitors, and in patients with concurrent conditions like lymphoma or certain vasculitides. Screen and manage hepatitis B (reactivation risk) before use.

BOARD TRAP

A board scenario of RA with a relative contraindication to TNF inhibitors (e.g., history of lymphoma or demyelinating disease) where the wrong answer is another TNF inhibitor. The discriminator: rituximab (B-cell depletion) is a preferred non-TNF biologic in such patients; still screen for HBV reactivation before giving it.

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10. Tocilizumab / sarilumab (IL-6)

WHAT YOU SEE

Tocilizumab/sarilumab shield on the right shelf

WHAT IT ENCODES

Tocilizumab and sarilumab are IL-6 RECEPTOR blockers, effective for RA refractory to other DMARDs and notable for working well as MONOTHERAPY (when MTX can't be combined). Because IL-6 drives the acute-phase response, they normalize CRP/ESR and can suppress fever — and they may cause GI perforation (caution with diverticulitis), neutropenia, and dyslipidemia.

BOARD TRAP

Using a normal CRP/temperature to exclude serious infection in a patient on an IL-6 inhibitor. The discriminator: IL-6 blockade BLUNTS CRP and fever, so infection signs are MASKED — a patient can be septic with a near-normal CRP and no fever; clinical suspicion must override reassuring inflammatory markers.

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11. Abatacept (T-cell co-stim)

WHAT YOU SEE

Abatacept vial on shelf

WHAT IT ENCODES

Abatacept is a CTLA-4–Ig fusion protein that blocks T-cell COSTIMULATION (it binds CD80/86 on antigen-presenting cells, preventing the CD28 'second signal' needed for T-cell activation). It is an option for inadequate response to csDMARDs or other biologics and is often considered relatively favorable in patients at higher infection risk.

BOARD TRAP

Mixing up abatacept's mechanism with cytokine blockade. The discriminator: abatacept works UPSTREAM by inhibiting T-cell costimulation (CTLA-4 Ig / CD80-86), distinct from TNF inhibitors, IL-6 blockers (tocilizumab), or B-cell depletion (rituximab). As with all biologics, screen for TB/HBV before starting.

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12. JAK inhibitors (tofacitinib, etc.)

WHAT YOU SEE

Tofacitinib/baricitinib/upadacitinib oral icons high on the ladder

WHAT IT ENCODES

JAK inhibitors — tofacitinib, baricitinib, upadacitinib — are ORAL targeted synthetic DMARDs (tsDMARDs) blocking intracellular Janus kinase signaling of multiple cytokines. They are used after csDMARD (and often biologic) failure and are highly effective, with oral dosing as an advantage over injectable biologics.

BOARD TRAP

Choosing a JAK inhibitor for an older patient with cardiovascular risk factors or VTE history. The discriminator: JAK inhibitors carry a BOXED WARNING (from the ORAL Surveillance trial) for serious infection, MAJOR ADVERSE CARDIOVASCULAR EVENTS (MACE), VENOUS THROMBOEMBOLISM, MALIGNANCY, and mortality — they should be used with caution in patients ≥65 or with CV/clot/cancer risk, often after a TNF inhibitor is tried.

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13. NSAIDs (symptom only)

WHAT YOU SEE

NSAIDs / less-effective oral icons up top

WHAT IT ENCODES

NSAIDs provide symptomatic relief of pain and stiffness in RA but DO NOT modify disease, slow progression, or prevent erosions. They are adjuncts only, used short-term alongside DMARDs, with attention to GI (ulcer/bleeding), renal, and cardiovascular toxicity.

BOARD TRAP

A newly diagnosed RA patient placed on an NSAID ALONE as definitive therapy. The discriminator: NSAIDs are NOT disease-modifying and never the sole treatment — a patient on NSAID monotherapy will continue to erode joints. Every RA patient needs a DMARD; NSAIDs only cover symptoms in the interim.

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14. Pre-treatment screening

WHAT YOU SEE

Anti-TNF shield with TB/hepatitis screening icons, exercise/rehab

WHAT IT ENCODES

Before starting biologics or JAK inhibitors: screen for latent TUBERCULOSIS (IGRA/PPD + CXR), HEPATITIS B and C, and bring vaccinations up to date — including pneumococcal, annual influenza, COVID, and (per risk) zoster. LIVE vaccines should be given BEFORE immunosuppression. Non-pharmacologic care (physical/occupational therapy, exercise, smoking cessation) complements drug therapy.

BOARD TRAP

Administering a LIVE vaccine (e.g., live zoster, MMR, intranasal influenza, yellow fever) to a patient already on a biologic/JAK inhibitor. The discriminator: LIVE vaccines are CONTRAINDICATED during biologic/immunosuppressive therapy (risk of disseminated infection) — they must be given beforehand; inactivated vaccines are safe during treatment. Vaccinating with a live vaccine on therapy is the trap.
